

Generative or Not?

Supplementary notes to think through the exercise

1 Why this exercise exists

The point isn't to test whether you can label things correctly. It's to make you notice that the word "AI" hides at least two very different kinds of systems, and that the failure modes of each are different. If you assume your spam filter and an LLM work the same way, you'll end up trusting each of them in the wrong places.

2 The core distinction in one paragraph

Discriminative / predictive AI learns to sort, rank, score, or classify existing inputs. It answers questions like *is this email spam?*, *how likely is this transaction to be fraud?*, *which video will this user click next?* The output is a label, a probability, or a ranking. The set of possible answers is fixed in advance.

Generative AI learns the distribution of its training data well enough to produce new samples from it. It answers *what comes next?*, *what would an image of this look like?*, *what is a plausible response to this prompt?* The output is content that didn't exist before you asked. The answer space is open-ended.

Quick heuristic. If the output is something that didn't exist before you asked, it's generative. If it's a label, score, or ordering applied to things that did exist, it's discriminative. Watch out: autocomplete *is* generative — it's producing tokens that aren't yet in your draft.

3 Step 1 scaffolding: where to look for AI in your week

You're more surrounded by AI than you probably realize. A non-exhaustive list to prime your search:

- **Communication.** Email spam filter, smart reply suggestions, phone keyboard autocomplete, grammar/spelling suggestions, translation.
- **Media.** Video recommendations (YouTube, TikTok), music recommendations (Spotify), news feed ranking, photo auto-tagging, search ranking.
- **Assistants.** Claude, ChatGPT, voice assistants (Siri, Alexa), summarization tools, AI image generators.
- **Background systems.** Face unlock, card-transaction fraud detection, captcha scoring, ad targeting, ETA prediction in Maps.

- **Productivity / coding.** Copilot, AI doc search, slide generators, meeting transcription, smart compose in email.
- **Physics-adjacent.** Jet tagging, anomaly detection in detector data, ML-driven trigger systems, generative simulation surrogates. (These blur the line — the model itself can be generative *or* discriminative depending on the task.)

4 Step 2 scaffolding: how to make the call

For each feature, ask three sub-questions:

1. **Is the output content or a label?** A sentence the model wrote is content. “0.97 probability of spam” is a label.
2. **Was the answer space fixed or open?** Spam vs. not-spam is fixed (two options). What an LLM can say is effectively unbounded.
3. **Could the same input produce many valid outputs?** If yes, that’s a signature of generation. If only one output is “correct,” it’s classification.

Common pitfalls.

- Autocomplete *feels* predictive but is generative.
- Recommender systems feel like “suggestions” but are discriminative — they rank items that already exist.
- Translation is generative (the target sentence is constructed, not retrieved).
- Search results are discriminative (ranking existing pages); an AI summary on top of search results is generative.

5 Step 3 scaffolding: judging the AI’s calibration

When you hand your list to an AI and ask which feature is most likely to have a failure mode *this course will help you understand*, judge its answer this way:

- Does it name a *specific* failure mode (e.g. “hallucinated facts in autocomplete”), or just a generic concern (“it might be wrong”)?
- Does the failure mode actually require a course on *generative* models to understand? If the same concern would apply to your spam filter, it didn’t really use the distinction.

6 Step 4: the four property questions, explained

These are predictions you’re making about which property of generative AI matters most for each task on your earlier list. The course revisits each property; for now, here’s a working definition.

6.1 Next Token Prediction — “Where do the answers come from?”

What it means. Generative LLMs produce text one token at a time, each token sampled from a probability distribution conditioned on everything so far. There is no separate fact-lookup step —

the model is sampling from learned patterns. This is why a confident-sounding wrong derivation can happen: each next token can be locally plausible without the whole answer being correct.

Tag with this when: the risk is “the answer might look right but is fabricated step-by-step.” Physics derivations, code, math, citations.

6.2 Knowledge — “What does it know?”

What it means. The model’s knowledge is whatever was captured in its training data, plus whatever it can retrieve or be shown in context. It has a cutoff date, gaps in specialized literature, and a bias toward popular sources. It doesn’t literally store facts — it stores patterns that often reproduce facts.

Tag with this when: success depends on the model actually knowing the right subfield, vocabulary, or recent literature. Your weekly exHEP paper search and the ML-for-HEP curriculum both lean heavily here.

6.3 Working Memory — “What’s it paying attention to?”

What it means. The model can only condition on what’s in its context window. Things you said three turns ago are visible; things in a file you never showed it are not. Long contexts have known attention-falloff effects — material in the middle of a long context tends to be weighted less.

Tag with this when: the task is multi-turn and the model must track constraints, prior decisions, or specific user details. The apartment moving planner is a classic working-memory task — if your constraints (lease dates, car, budget) aren’t kept in context, the model will reinvent them.

6.4 Steerability — “How much am I in control?”

What it means. How reliably can you push the model toward the behavior you want? Prompts, examples, output-format constraints, system prompts, and structured outputs all increase steerability. Vague prompts cede control to the model’s defaults.

Tag with this when: the risk is “I’ll get a reasonable-looking output that isn’t what I actually wanted.” Especially relevant for the “teach me like a grad student” task — pedagogical style is the thing you most want to control.

7 Tags for your four tasks (a starting point to push back on)

You don’t have to agree with these. The point is to have a prediction on paper.

- **Task 1 — Nuclear/particle physics, grad-student style.** Primary: *Next Token Prediction* (confident wrong derivations). Secondary: *Steerability* (pedagogical style).
- **Task 2 — ML/AI course for exHEP.** Primary: *Knowledge* (does it actually know HEP-specific workflows?). Secondary: *Steerability* (forcing it past generic ML).
- **Task 3 — Weekly exHEP paper search.** Primary: *Knowledge* (arXiv-category coverage, training cutoffs). Secondary: *NTP* (fabricated titles/authors).
- **Task 4 — Apartment moving planner.** Primary: *Working Memory* (holding your constraints). Secondary: *Steerability* (making it ask before assuming).

8 Worksheet

Your five AI features this week:

AI feature	Your call (G/D)	Why you think so
1.		
2.		
3.		
4.		
5.		

Most likely failure mode this course will help with:

Your tag prediction for the four earlier tasks:

- Task 1 (physics teaching): _____
- Task 2 (ML/AI exHEP course): _____
- Task 3 (weekly paper search): _____
- Task 4 (moving planner): _____

9 Do this in order

1. Spend 5 minutes listing your five AI features. Don't overthink.
2. For each, write G or D and a one-line reason.
3. Hand the list to an AI; ask it to check you. Note which ones it disagrees with you on and why.
4. For the four tasks from earlier, write your tag prediction.
5. Save this sheet. Revisit after each of the next four lessons and see whether your tags still feel right — or whether the course shifted them.

You are not expected to get any of this right on the first pass. You are making predictions you will test.